



10778 - Star-Disk Wars: Revealing the dominant outflow-launching mechanism in partially mapped Class II disks

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRSpec IFU Spectroscopy	(1) HV-TAU-C
	2		NIRSpec IFU Spectroscopy	(2) OPH-163131
	3		NIRSpec IFU Spectroscopy	(3) FLYING-SAUCER
	4		NIRSpec IFU Spectroscopy	(4) ESO-HA-574
	5		NIRSpec IFU Spectroscopy	(5) LK-HA-263C
	6		NIRSpec IFU Spectroscopy	(6) HK-TAU-B
	8		NIRSpec IFU Spectroscopy	(8) HH-48-NE

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
	9		NIRSpec IFU Spectroscopy	(9) ESO-HA-569

ABSTRACT

Jets and winds play a central role in protoplanetary disk evolution and thus in planet formation. Though widely accepted as magnetically driven, it remains unclear whether their launch is dominated by magnetospheric ejections (ME) or radially extended MHD disk winds. A recent breakthrough revealed narrow forbidden atomic-line emission and broad, hollow, cone-like rovibrational H₂. The origin of this H₂ emission is directly linked to the dominant launching mechanism. If ME dominates, then H₂ would trace the surface of a shock-excited cavity. In that case, rovibrational and rotational H₂ lines will trace similar temperatures and opening angles, and we expect a strong correlation between jet-shock parameters and H₂ strength. Conversely, if MHD disk winds dominate, then the H₂ emission should arise in a disk wind, creating nested morphologies of multiple H₂ lines and breaking the correlation between shock parameters and H₂ properties.

To test this, we propose NIRSpec IFU 1-1.9 micron observations to trace atomic jets towards eight edge-on disks with cone-like emission detected in H₂ rovibrational (archival NIRSpec 1.7-5.3 micron) and pure rotational lines (archival MIRI, 5-28 micron). For each line, we will measure the opening angle and assess whether rovibrational H₂ is nested inside rotational H₂, and if [Fe II] is the narrowest. We will also derive shock velocities and pre-shock jet densities using diagnostic line ratios and compare them to H₂ morphology and flux. By resolving the nested H₂ emission and linking jet shocks to H₂ morphology, we will make a transformative step toward revealing the dominant outflow-launching mechanism in the Class II stage of disk evolution.

OBSERVING DESCRIPTION

We aim to spatially resolve jets towards eight Class II edge-on disks. For this goal, we will use NIRSpec IFU and target ~50 forbidden atomic lines, including [Fe II] 1.257, 1.533, 1.644 micron, [S II] 1.03 micron and [N I] 1.04 micron lines. As the selected lines fall in the G140H/F100LP disperser-filter combination settings and the longer wavelength spectro-imaging data (G235H/F170LP, G395H/F290LP, MIRI MRS) already exists for most disks, we will obtain 1-1.9 micron data for all disks and 1.7-3.2 micron data for two disks that lack archival observation. The exposure settings are set to attain S/N > 3 per pixel nearly 1 arcsec away from the center in the weak [Fe II] 1.533 micron line, using conservative line flux estimates. With this exposure, we will achieve S/N > 10 per pixel in the [Fe II] 1.644 micron line and a total line SNR > 30. As the continuum emission is dominated by an extended edge-on disk, we do not propose for target acquisition. We select 4-Point dithering to enhance pixel sampling and mitigate the effects of undersampling. We request a duplicate observation of HK TauB as the archival observation (GO 1621, PI:I. Pascucci) is off-centered leading to saturation of the entire cube from much brighter companion HK TauA.

Proposal 10778 - Targets - Star-Disk Wars: Revealing the dominant outflow-launching mechanism in partially mapped Class II disks

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	HV-TAU-C	RA: 04 38 35.5126 (69.6479692d) Dec: +26 10 41.32 (26.17814d) Equinox: J2000	Proper Motion RA: 4.775979175959282 mas/yr Proper Motion Dec: -21.376969915794696 mas/yr Epoch of Position: 2000	
	Comments: Category=Star Description=[Pre-main sequence stars, Protoplanetary disks, Protostars] Extended=YES				
	(2)	OPH-163131	RA: 16 31 31.2306 (247.8801275d) Dec: -24 26 28.59 (-24.44128d) Equinox: J2000	Epoch of Position: 2024.7	
	Comments: Based on archival observations, the coordinates are set to keep the spatially resolved dark lane at the center. Category=Star Description=[Pre-main sequence stars, Protoplanetary disks, Protostars] Extended=YES				
	(3)	FLYING-SAUCER	RA: 16 28 13.6800 (247.0570000d) Dec: -24 31 39.83 (-24.52773d) Equinox: J2000	Epoch of Position: 2024.8	
	Comments: Based on archival observations, the coordinates are set to keep the spatially resolved 'dark lane' at the center of the FOV Category=Star Description=[Pre-main sequence stars, Protoplanetary disks, Protostars] Extended=YES				
	(4)	ESO-HA-574	RA: 11 16 2.7639 (169.0115162d) Dec: -76 24 53.24 (-76.41479d) Equinox: J2000	Proper Motion RA: -23.103 mas/yr Proper Motion Dec: -2.9440000389513443 mas/yr Epoch of Position: 2000	
	Comments: Category=Star Description=[Pre-main sequence stars, Protoplanetary disks, Protostars] Extended=YES				
	(5)	LK-HA-263C	RA: 02 56 8.6700 (44.0361250d) Dec: +20 03 40.60 (20.06128d) Equinox: J2000	Proper Motion RA: 2.6 mas/yr Proper Motion Dec: -2.3999999939405825 mas/yr Epoch of Position: 2000	
	Comments: Category=Star Description=[Pre-main sequence stars, Protoplanetary disks, Protostars] Extended=YES				
	(6)	HK-TAU-B	RA: 04 31 50.6100 (67.9608750d) Dec: +24 24 14.50 (24.40403d) Equinox: J2000	Epoch of Position: 2022.75	
	Comments: The RA/DEC is decided based on archival HKTau observations to avoid HK Tau A in the FOV. The coordinates are slightly offset from HK Tau B. Category=Star Description=[Pre-main sequence stars, Protoplanetary disks, Protostars] Extended=YES				
	(8)	HH-48-NE	RA: 11 04 23.3100 (166.0971250d) Dec: -77 18 6.70 (-77.30186d) Equinox: J2000	Epoch of Position: 2022.6	
	Comments: Based on archival observations, the coordinates are set to keep HH48NE as close to the center as possible while excluding HH48SW from the FOV. Category=Star Description=[Pre-main sequence stars, Protoplanetary disks, Protostars] Extended=YES				

Proposal 10778 - Targets - Star-Disk Wars: Revealing the dominant outflow-launching mechanism in partially mapped Class II disks

(9)	ESO-HA-569	RA: 11 11 10.7144 (167.7946433d) Dec: -76 41 57.43 (-76.69929d) Equinox: J2000	Epoch of Position: 2023.2
<i>Comments: Based on archival observations, the coordinates are set to keep the spatially resolved 'dark lane' at the center of the FOV.</i> <i>Category=Star</i> <i>Description=[Pre-main sequence stars, Protoplanetary disks, Protostars]</i> <i>Extended=YES</i>			

Proposal 10778 - Observation 1 - Star-Disk Wars: Revealing the dominant outflow-launching mechanism in partially mapped Class II d...

Observation	Proposal 10778, Observation 1											Wed Apr 29 21:00:24 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	HV-TAU-C	RA: 04 38 35.5126 (69.6479692d)				Proper Motion RA: 4.775979175959282 mas/yr					
			Dec: +26 10 41.32 (26.17814d)				Proper Motion Dec: -21.376969915794696 mas/yr					
			Equinox: J2000				Epoch of Position: 2000					
			Comments: Category=Star Description=[Pre-main sequence stars, Protoplanetary disks, Protostars] Extended=YES									
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140H/F100LP	NRSIRS2RAPID	40	1	false	true	NONE	4	4	2392.578	

Proposal 10778 - Observation 2 - Star-Disk Wars: Revealing the dominant outflow-launching mechanism in partially mapped Class II d...

Observation	Proposal 10778, Observation 2											Wed Apr 29 21:00:24 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSspec IFU Spectroscopy											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(2)	OPH-163131	RA: 16 31 31.2306 (247.8801275d)				Epoch of Position: 2024.7					
			Dec: -24 26 28.59 (-24.44128d)									
			Equinox: J2000									
			Comments: Based on archival observations, the coordinates are set to keep the spatially resolved dark lane at the center.									
Template	Category=Star											
	Description=[Pre-main sequence stars, Protoplanetary disks, Protostars]											
	Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140H/F100LP	NRSIRS2RAPID	40	1	false	true	NONE	4	4	2392.578	

Proposal 10778 - Observation 3 - Star-Disk Wars: Revealing the dominant outflow-launching mechanism in partially mapped Class II d...

Observation	Proposal 10778, Observation 3											Wed Apr 29 21:00:24 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(3)	FLYING-SAUCE	RA: 16 28 13.6800 (247.0570000d)				Epoch of Position: 2024.8					
			Dec: -24 31 39.83 (-24.52773d)									
			Equinox: J2000									
			Comments: Based on archival observations, the coordinates are set to keep the spatially resolved 'dark lane' at the center of the FOV									
Template	Category=Star											
	Description=[Pre-main sequence stars, Protoplanetary disks, Protostars]											
	Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140H/F100LP	NRSIRS2RAPID	40	1	false	true	NONE	4	4	2392.578	

Proposal 10778 - Observation 4 - Star-Disk Wars: Revealing the dominant outflow-launching mechanism in partially mapped Class II d...

Observation	Proposal 10778, Observation 4											Wed Apr 29 21:00:24 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(4)	ESO-HA-574	RA: 11 16 2.7639 (169.0115162d) Dec: -76 24 53.24 (-76.41479d) Equinox: J2000				Proper Motion RA: -23.103 mas/yr Proper Motion Dec: -2.9440000389513443 mas/yr Epoch of Position: 2000					
	Comments: Category=Star Description=[Pre-main sequence stars, Protoplanetary disks, Protostars] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140H/F100LP	NRSIRS2RAPID	40	1	false	true	NONE	4	4	2392.578	

Proposal 10778 - Observation 5 - Star-Disk Wars: Revealing the dominant outflow-launching mechanism in partially mapped Class II d...

Observation	Proposal 10778, Observation 5											Wed Apr 29 21:00:24 GMT 2026
	Diagnostic Status: Warning											
Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(5)	LK-HA-263C	RA: 02 56 8.6700 (44.0361250d)				Proper Motion RA: 2.6 mas/yr					
	Dec: +20 03 40.60 (20.06128d)				Proper Motion Dec: -2.3999999939405825 mas/yr							
	Equinox: J2000				Epoch of Position: 2000							
	Comments: Category=Star Description=[Pre-main sequence stars, Protoplanetary disks, Protostars] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140H/F100LP	NRSIRS2RAPID	40	1	false	true	NONE	4	4	2392.578	

Proposal 10778 - Observation 6 - Star-Disk Wars: Revealing the dominant outflow-launching mechanism in partially mapped Class II d...

Observation	Proposal 10778, Observation 6											Wed Apr 29 21:00:24 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(6)	HK-TAU-B	RA: 04 31 50.6100 (67.9608750d) Dec: +24 24 14.50 (24.40403d) Equinox: J2000				Epoch of Position: 2022.75					
	Comments: The RA/DEC is decided based on archival HKTau observations to avoid HK Tau A in the FOV. The coordinates are slightly offset from HK Tau B.											
	Category=Star Description=[Pre-main sequence stars, Protoplanetary disks, Protostars] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140H/F100LP	NRSIRS2RAPID	40	1	false	true	NONE	4	4	2392.578	
	2	G235H/F170LP	NRSIRS2RAPID	40	1	false	true	NONE	4	4	2392.578	

Proposal 10778 - Observation 8 - Star-Disk Wars: Revealing the dominant outflow-launching mechanism in partially mapped Class II d...

Observation	Proposal 10778, Observation 8											Wed Apr 29 21:00:24 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(8)	HH-48-NE	RA: 11 04 23.3100 (166.0971250d) Dec: -77 18 6.70 (-77.30186d) Equinox: J2000				Epoch of Position: 2022.6					
	Comments: Based on archival observations, the coordinates are set to keep HH48NE as close to the center as possible while excluding HH48SW from the FOV.											
	Category=Star											
	Description=[Pre-main sequence stars, Protoplanetary disks, Protostars] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140H/F100LP	NRSIRS2RAPID	40	1	false	true	NONE	4	4	2392.578	
	2	G235H/F170LP	NRSIRS2RAPID	40	1	false	true	NONE	4	4	2392.578	

Proposal 10778 - Observation 9 - Star-Disk Wars: Revealing the dominant outflow-launching mechanism in partially mapped Class II d...

Observation	Proposal 10778, Observation 9											Wed Apr 29 21:00:24 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSspec IFU Spectroscopy											
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(9)	ESO-HA-569	RA: 11 11 10.7144 (167.7946433d) Dec: -76 41 57.43 (-76.69929d) Equinox: J2000			Epoch of Position: 2023.2						
	Comments: Based on archival observations, the coordinates are set to keep the spatially resolved 'dark lane' at the center of the FOV. Category=Star Description=[Pre-main sequence stars, Protoplanetary disks, Protostars] Extended=YES											
Template	TA Method					HFF Readout Mode						
	NONE					false						
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140H/F100LP	NRSIRS2RAPID	40	1	false	true	NONE	4	4	2392.578	